Yellowfin Model Development

First Model Runs and Ideas to Explore

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Model development

Pacific Community
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Spatial structure

1. Consider single-region or equatorial model
2. Simplify regional structure for bigeye

Movement

3. SEAPODYM coefficients

Fisheries

4. Convert weight comps to length comps
5. Combine fisheries
6. Lower effective sample size in general
7. Analyze data conflict, identify and test different approaches
8. Possibly reduce sample size in regions without otolith data
9. Consider different scenarios for Indonesian catches

Natural mortality

10. Different levels of M
    (continuous or discrete)

Growth

11. Different levels of k
    (continuous or discrete)
12. Richards curve

Selectivity

13. Length-based

Tags

14. Mixing period for each release group

CPUE

15. Alternative indices
16. Time-varying CV

These slides:

https://github.com/PacificCommunity/ofp-sam-yft-2026-assessment